



Five invoices, one month-end, and none of the data leaves.

The presenter's cheat sheet for a six-minute slot: two minutes of slides, three minutes live, one minute to close. It covers what to pre-stage, which invoice to show when, what appears on screen and what to say. Every score below was produced by running the sample PDFs through the app's own parser and rules engine against the seeded history.

6 min slot

5 sample invoices + encore

73 seeded invoices · 9 vendors

Local model: qwen2.5 3B on Ollama

InvoiceShield checks every invoice before the money moves, and the invoice, the model and the decision never leave the company.

HOW IT WORKS

From PDF to decision in seven steps

Rules find the problems. The model only reads fields and explains. A person makes every decision, and every step is logged.

1	2	3	4	5	6	7
Upload	Read	LOCAL AI Extract	Check	Score	LOCAL AI Explain	HUMAN Review
PDF, PNG or JPEG. Type checked by magic bytes; SHA-256 hash stored.	Text layer via unpdf, or Tesseract OCR for images.	Vendor, number, date, amounts, IBAN last 4. Numbers kept only if they appear in the document.	Nine deterministic rules against the vendor register and invoice history.	High 40 · Medium 18 · Low 6, capped at 100.	Plain-English summary and verification steps. Verdict words blocked.	Mark reviewed, flag, clear, reopen, or correct fields. Append-only audit trail.

Score levels: **Low 0-14** **Medium 15-39** **High 40-69** **Critical 70+**

Rules find problems, AI doesn't

Every finding comes from a tested check and carries its evidence: this value against that value.

Nothing leaves the machine

With Ollama, invoice text stays local. With no model at all, the app runs on rules and template summaries.

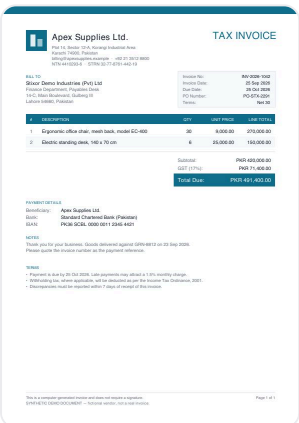
THE CAST

The sample invoices

In `public/samples/`, also listed under **Try a synthetic sample** on the upload page. Upload them in this order, after loading the demo data. Click a thumbnail to enlarge.

The model can't invent numbers

An amount from the model is rejected unless its digits are in the document. The parser value is used and the field is marked uncertain.



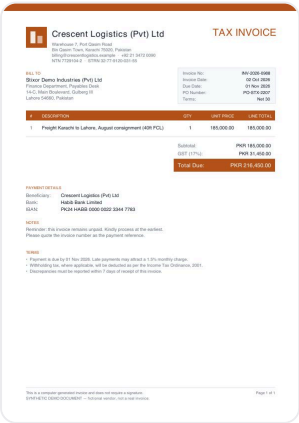
1 • The normal one

Low 0

01-legitimate-apex-supplies.pdf · Apex Supplies Ltd. · INV-2026-1042 · PKR 491,400

A routine furniture order from a trusted supplier. Registered vendor, registered IBAN, amount in range, and the maths adds up.

No findings. This is what most invoices look like.



2 • The reminder

High 40

02-duplicate-crescent-logistics.pdf · Crescent Logistics (Pvt) Ltd · INV-2026-0988 · PKR 216,450

"Reminder: this invoice remains unpaid." The ledger already holds this number from this vendor, twice.

HIGH Invoice number already recorded for this vendor
2 earlier records · most recent PKR 157,716 dated 28 Sep



High 58

"Please note our updated banking details." A real vendor, a new IBAN, and far more than Meridian usually bills.

MEDIUM Amount outside the vendor's typical range
PKR 2,340,000 · typical PKR 200,000 – 1,500,000



Medium 36

One letter off a registered supplier, a different address, a mobile number, "urgent: release payment today". The total carries PKR 20,000 that no line item explains.

MEDIUM Subtotal + tax does not equal total
Expected PKR 175,500 · stated PKR 195,500 · difference PKR 20,000



Critical 98

A previously processed invoice number, replayed with new bank details, the amount roughly quadrupled, due in two days. The pattern of a compromised vendor mailbox.

MEDIUM Amount outside the vendor's typical range
PKR 1,450,800 · typical PKR 120,000 – 500,000

HIGH Invoice number already recorded for this vendor
INV-2026-1042, same total

10 MINUTES BEFORE

Pre-stage before your slot

Ticks are saved in this browser only.

- ☐ **Model warm**
`ollama serve`, then `ollama run qwen2.5:3b "hi"` once. Keep a terminal with `ollama ps` ready for the proof moment.
- ☐ **App running**
`npm run dev`, or `npm run emulators` + `npm run dev:emulator` for a fully local stack.
- ☐ **Fresh account, then Load synthetic demo data**
The toast should read "Loaded 9 vendors and 73 synthetic invoices".
- ☐ **Pre-stage invoices 2, 3 and 4**
Check they show High 40, High 58 and Medium 36. Do not upload 1 or 5 yet.
- ☐ **Tabs ready: deck, app Overview, app Upload invoice**
- ☐ **Browser zoom 110–125%, notifications off**

The script

Left: what to click. Right: what to say. Invoices 2, 3 and 4 are pre-staged; 1 and 5 are uploaded live. Say it in your own words; the beats matter more than the lines.

0:00 · 2 min

Slides 1–6

Deck

SHOW

- **Cover:** who and what.
- **Problem:** three invoices that look normal.
- **Why sovereign:** the theme.
- **What we built, Architecture, Role of AI.**

SAY

The invoices that cost money don't look wrong: a duplicate sent as a reminder, a real supplier whose bank details changed, a name one letter off. And an invoice is bank accounts, pricing and payment timing, the data you can't send to a cloud AI. So the AI comes to the data.

Everything in this dashed line runs on this laptop right now. The AI does the reading; it can't invent a number or give a verdict, and the app still works if it's off.

2:00 · 20 s

① Overview

App

DO

- KPIs and risk mix.
- Review queue led by the pre-staged Meridian, Crescent and "Apex Supplies".

SAY

Month-end at Stixor Demo Industries: 73 invoices processed, most of them green. The queue is sorted by risk, so the first thing you see is what's most likely to cost money.

2:20 · 40 s

② Invoice

1, live

Upload invoice

The normal one Low 0

DO

- Try a **synthetic sample** → **Legitimate invoice**.
- Narrate the stages as they tick.
- Show **Extracted fields**, then **Mark reviewed**.

SAY

The local model read vendor, number, date, amounts and IBAN straight off the PDF. Registered vendor, registered account, the maths adds up. That's most invoices: they pass quietly.

3:00 · 40 s

③ Meridian

Pre-staged

The quiet bank change High 58

DO

- Point at 8810 vs 5532 and the amount range.
- Read the **Summary** and first verification step.
- **Flag for investigation**, note: "Call Meridian on the number in the vendor master to confirm the IBAN change."

SAY

Real vendor, friendly note, new IBAN. The summary was written by the model on this laptop. It doesn't say fraud, because it isn't allowed to; it says what to check. The note is mandatory, so the next person knows what's pending.

3:40 · 30 s

④ "Apex Suplies"

Pre-staged

The impostor Medium 36 → after correction Medium 18

DO

- Let the room spot the missing "p" (92% similar).
- Set Total to → **Save and re-run checks**.

SAY

One letter. A person misses it at 5pm; a string comparison doesn't. If the vendor sends a corrected total, I fix it, the checks re-run instantly, and the correction is logged.

4:10 · 40 s

The replay Critical 98

⑤ Invoice

5, live

Upload invoice · the climax

DO

- **Try a synthetic sample → Replayed invoice, new bank.**
- Pause on the three stacked findings.

SAY

This is what someone inside a vendor's mailbox does: replay a real invoice, swap the bank details, inflate the amount, make it due in two days. Any one might slip through. Together: Critical. This one never gets paid without a phone call.

4:50 · 25 s

⑥ Audit and

⑦ proof

Audit trail · terminal

DO

- **Audit trail:** uploaded → analysed → reviewed → flagged → corrected → analysed again.
- Terminal: `ollama ps` shows `qwen2.5:3b` loaded locally.

SAY

Append-only: who, what, when, before and after, and why. And that's the model. It's on this machine.

5:15 · 1 min

Slides 8–11

Deck

SHOW

- **What you just saw, Impact, Technical choices** (if technical room), **Close.**
- Encore only if time: upload invoice 1 again → Critical 80.

SAY

9 of 73 invoices raised a finding; the rest pass quietly. Zero bytes to an external AI. Every decision recorded.

InvoiceShield doesn't approve, reject or pay anything. It reads every invoice, checks it against everything the company knows, and shows the evidence, without the data ever leaving. Thank you.

Likely questions

QUESTION	ANSWER
Where's the AI beyond a wrapper?	The local model turns unstructured PDFs and scans into fields and writes the explanation, inside guardrails: numbers checked against the document, verdict words blocked, findings only from rules, rules-only fallback.
What's sovereign about it?	Model runs locally on Ollama, files are on local disk, no external AI calls. Data store on the Firebase emulators on-prem, or the company's own project. Be precise if asked: the default config keeps auth and metadata in a Firebase project; emulator mode is fully local.
What data did you use?	Entirely synthetic: fictional vendors and generated invoices, marked as synthetic on every PDF.
Is the score a fraud probability?	No. It ranks what to review first, built from rule severities. The findings beside it are what matter.
What if extraction is wrong?	Fields it couldn't read reliably are called out. The reviewer corrects them and re-runs checks; the correction is audited.
Scanned invoices?	Images go through Tesseract OCR. A scanned PDF with no text layer is detected and the user is asked to upload an image.
Can it pay or block invoices?	Deliberately not. It's a review step before your ERP's payment run.
How does it know a vendor's normal amount?	A configured range per vendor, plus a robust z-score (median and MAD) against that vendor's history once there are 4+ invoices.
Teams and approvals?	The prototype is per-user. Organisation workspaces, roles and ERP integration are on the roadmap.

Live recovery

SYMPTOM	FIX
No seed button, or it errors	The account already has data. Sign up a new account.
Sample 2 or 5 shows no duplicate	Demo data wasn't loaded first, or went into a different account.
Summary source says template	The model is down or timed out. Say "this is the rules-only fallback, it degrades gracefully" and carry on.
Stuck on Extracting	Model cold start. Wait about 20 seconds, or set <code>AI_PROVIDER=none</code> before the demo.
"Identical file already uploaded" too early	That sample was already uploaded in this account. Use a fresh one, or make it the encore.

All vendors, people and invoices are fictional and marked as synthetic. Regenerate the PDFs with `npm run samples`; the full script is also in `DEMO.md` and the submission write-up in `SUBMISSION.md`. Slides: [InvoiceShield Hackathon Pitch](#).